

Track-Specific Week 6 Output

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1. Week 5 → Week 6 Transition

- Main Week 5 output: Initial HealthConnect Analytics Report — data preparation, EDA across 5 business questions, 5 KPIs, and a Power BI dashboard with KPI cards, 5 charts, and a slicer.
- Most important result: Previous no-show history was the strongest predictor of future no-shows (43.5% to 68.8% as prior no-shows increase), followed by distance to the clinic (46.5% to 57.8%).
- Main issue/limitation discovered: The waiting-time analysis required a derived calculation linking a patient's current visit to their next visit's outcome, which was technically difficult to build correctly in Power BI and needed independent verification.
- Relevant track for Week 6: Data Science — the strongest predictors identified (previous_no_shows, distance_to_clinic_km) are directly useful as candidate features for a no-show prediction model.
- Focus for Week 6: Validate whether the previous no-show and distance findings hold up across patient segments (age group, appointment type), test their combined effect, and package the findings as concrete feature recommendations for model development.

2. Advanced Analysis — Validating Week 5's Strongest Findings

2.1 Does the Previous No-Show Effect Hold Across Age Groups?

The Week 5 finding (no-show rate rising sharply with prior no-show count) was tested separately within each age group to confirm it is not being driven by one age segment alone.

- 18-24: 45.2% (0 prior) -> 56.9% (1-2) -> 84.6% (3+)
- 25-34: 47.3% -> 53.9% -> 77.8%
- 35-44: 44.4% -> 53.3% -> 70.0%
- 45-54: 44.3% -> 53.2% -> 57.1%
- 55-64: 46.0% -> 56.1% -> 81.8%
- 65+: 37.7% -> 55.7% -> 55.6%

Finding: The pattern holds across every age group without exception, confirming this is a genuine, broad-based behavioural signal rather than an artefact of one segment. The effect is most extreme in patients aged 18-24 and 55-64, where a 3+ no-show history is associated with over 80% likelihood of missing the next appointment.

2.2 Does the Distance Effect Hold Across Appointment Types?

- Follow-up: 51.7% (0-5km) rising to 56.4% (20km+)
- Specialist Consultation: 47.6% rising to 56.0%
- General Consultation: 44.7% rising to 59.5%
- Diagnostic Test: 39.4% rising to 61.7% (15-20km), the steepest increase of any appointment type

Finding: The distance effect holds across all four appointment types, but is strongest for Diagnostic Tests, where no-show rate more than doubles between the nearest and farthest patient groups. This suggests diagnostic appointments, which may be perceived as less urgent than consultations, are especially sensitive to travel burden.

2.3 Combined Effect: Previous No-Shows x Distance

Testing both factors together revealed an important refinement to the Week 5 findings, rather than a simple additive effect.

- Patients with 0 previous no-shows: rate climbs steadily with distance, from 42.1% (0-5km) to 53.0% (20km+) — distance matters most for this group.
- Patients with 1-2 previous no-shows: rate climbs from 51.4% to 64.0% across the same distance range.
- Patients with 3+ previous no-shows: rate is already very high at short distances (89.5% at 0-5km) and does not continue climbing with distance — it actually plateaus and slightly declines at longer distances.

Finding: Distance meaningfully compounds risk for patients with low or moderate prior no-show history, but for patients who already have a strong history of missing appointments (3+), distance stops being the dominant factor — their risk is already near-ceiling regardless of location. This refines the Week 5 conclusion: distance and history are not independent, additive risks; history dominates once it reaches a high threshold.

3. KPI Validation

The core No-Show Rate KPI (48.5%) was tested for stability across segments not previously examined in Week 5.

- By Appointment Type: Diagnostic Test 49.7%, Follow-up 51.2%, General Consultation 46.6%, Specialist Consultation 47.4% — reasonably stable, with Follow-up appointments slightly elevated.
- By Age Group: ranges from 45.1% (65+) to 50.7% (25-34 and 55-64) — stable, with older patients (65+) showing a modestly lower overall no-show rate.

Conclusion: The overall 48.5% No-Show Rate KPI is validated as a reliable, representative figure — it does not appear to be skewed by any single segment, confirming it is appropriate to use as the clinic's headline metric.

4. Updated Dashboard / Visualisation

The Week 5 Power BI dashboard was extended, not rebuilt, with two additional views based on the Week 6 findings: a combined Previous No-Shows x Distance matrix visual, and a No-Show Rate by

Appointment Type chart. These additions directly support the refined finding that previous no-show history dominates risk at high levels, while distance is the more useful signal for lower-risk patients.

5. Cross-Track Integration

A. Integration Point

- Track collaborated with: Data Science (no-show prediction model).
- Project dependency: The Data Science track's Week 5 baseline model requires validated, high-signal features to prioritise during feature refinement in its own Week 6 work.
- Output received: The Data Science track's Week 5 problem definition — the model is a binary classifier predicting No-Show vs. Attended.
- Output provided: A validated, ranked feature recommendation based on Week 6 analysis (below).

B. Integration Objective

Before this integration, the Data Science track had a candidate feature list based only on assumption. After this integration, the model's feature prioritisation is evidence-based, using validated relationships confirmed across multiple patient segments rather than a single untested pass.

C. Evidence — Feature Recommendation for the No-Show Prediction Model

- `previous_no_shows` (or a grouped version: 0 / 1-2 / 3+): Highest priority. Confirmed as the strongest, most consistent predictor across every age group tested, with no exceptions.
- `distance_to_clinic_km`: Second priority, but recommended as an interaction feature with `previous_no_shows` rather than a standalone feature, since its effect is strongest for patients with low-to-moderate prior no-show history and weakens at high-risk levels. A combined feature (e.g., `distance x prior_no_show_flag`) may outperform using either variable alone.
- `appointment_type`: Recommended as a supporting categorical feature, since Diagnostic Test appointments show a distinctly steeper distance sensitivity than other types.
- `reminder_sent`: Recommended as a lower-priority feature; it showed only a modest standalone effect (4-point difference) in Week 5 and is unlikely to add strong predictive power on its own.
- `waiting_time_minutes`: Not recommended as a feature for predicting future no-shows, based on Week 5 findings showing no meaningful relationship with next-visit attendance.

What changed as a result: The candidate feature list for the prediction model is now ranked and evidence-based rather than assumed, with a specific recommendation to test an interaction feature (`distance x prior history`) rather than treating the two as independent inputs.

6. Business Recommendations

- Prioritise a patient-risk flagging system based primarily on previous no-show count, since this is the single strongest and most consistent signal across all patient segments tested.
- For patients with 0-2 previous no-shows, distance-based interventions (e.g., telehealth offers, transport support) are likely to be effective and should be targeted at this group specifically, since distance is not yet the dominant risk factor for them.

- For patients with 3+ previous no-shows, distance-based interventions are unlikely to move the needle significantly — direct, personal outreach (phone calls, confirmation calls) is more appropriate given their already very high baseline risk.
- Diagnostic Test appointments warrant a distance-specific review, given the unusually steep no-show increase for patients travelling further for this appointment type specifically.

7. Data Limitations

- The dataset remains fictional, so findings may not fully reflect real-world patient behaviour.
- Some segment combinations (e.g., 3+ previous no-shows at certain distance bands) have smaller sample sizes, which reduces confidence in the exact percentages for those specific combinations, though the overall direction of the pattern remains consistent.

8. Assumptions, Limitations, Risks and Dependencies (Updated)

- Resolved: The Week 5 concern about the technical reliability of the derived waiting-time calculation was addressed through independent verification; the finding (no meaningful effect) is confirmed stable and is not revisited further in Week 6.
- Still relevant: Dataset remains fictional; findings should be treated as illustrative rather than clinically validated.
- New issue discovered: The relationship between previous no-shows and distance is not simply additive — this was not apparent from the Week 5 single-variable analysis and required Week 6's combined analysis to detect.
- New dependency: The Data Science track's model performance now depends on correctly implementing the recommended interaction feature (distance x prior history), not just including both variables independently.
- Recommended mitigation: The Data Science track should test both an additive and an interaction-based feature approach during model development, comparing performance to confirm which structure works best.

9. Week 7 Analytical Testing Requirements

For Week 7, the analytical work should focus on testing whether the refined findings (particularly the previous-no-shows x distance interaction) translate into actual improved model performance once implemented by the Data Science track, and whether the appointment-type-specific distance sensitivity (Diagnostic Tests) holds up under further scrutiny.